Week 5 – Extended Kalman Filter Localization

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1 Introduction

The goal of this assignment was to implement a localization algorithm based on the Extended Kalman Filter (EKF) using GNSS measurements. The EKF was used to estimate the robot state, including position and orientation, despite nonlinear motion dynamics and partial observability.

2 Preparation

The `outdoor_1` map was used for this task. A trajectory was manually designed from the starting pose $[2, 2, \pi/2]$ to the goal position $[16, 2]$. The path consisted of multiple waypoints to ensure smooth navigation.

To initialize the filter, GNSS measurements were collected while the robot remained stationary. From these samples, the mean position and covariance matrix were computed. These values were later used to define the measurement noise covariance matrix.

3 EKF Prediction

The prediction step was implemented using the nonlinear motion model of a differential drive robot. The state vector is defined as $x = [x, y, \theta]^T$.

The motion model is given by:

$$x_{k+1} = x_k + v\cos(\theta_k)\Delta t, \quad y_{k+1} = y_k + v\sin(\theta_k)\Delta t, \quad \theta_{k+1} = \theta_k + \omega\Delta t$$

where v and ω are computed from the wheel velocities.

Due to the nonlinearity, the covariance is propagated using the Jacobian:

$$\Sigma^- = G\Sigma G^T + R$$

The process noise covariance was initialized as $R = \mathrm{diag}(0.01, 0.01, 0.01)$ and later tuned to balance smoothness and responsiveness.

4 Measurement Update

The correction step uses a linear measurement model, since GNSS provides only position measurements $z = [x, y]^T$.

The measurement matrix is:

$$C = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

The measurement covariance matrix Q was obtained from GNSS data collected during initialization.

The Kalman gain is computed as:

$$K = \Sigma^- C^T (C\Sigma^- C^T + Q)^{-1}$$

and used to update the state and covariance.

5 Filter Tuning with Known Initial Pose

In the first experiment, the initial pose was set to the true value $x_0 = [2, 2, \pi/2]$ with zero covariance, representing high certainty.

The EKF was able to track the robot trajectory accurately. The estimated position closely followed the true position, while the orientation was smoothly inferred from the motion model.

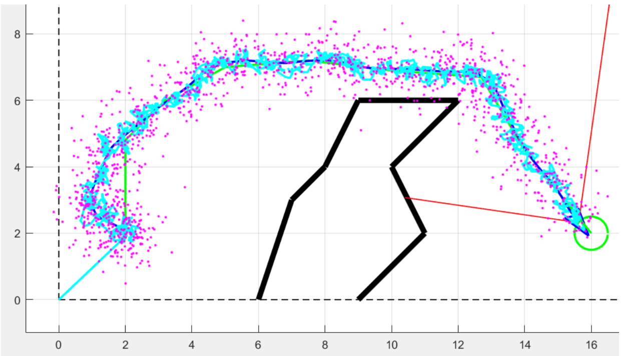

Figure 1: EKF estimation with known initial pose

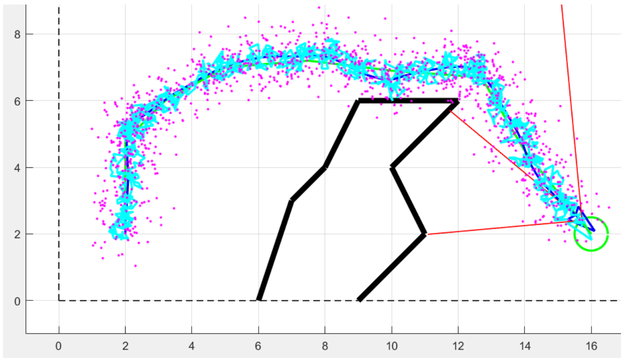

Figure 2: EKF convergence from uncertain initial pose

6 Localization with Unknown Initial Pose

In a more realistic scenario, the initial pose was not known. The initial position was estimated from GNSS measurements, while a high variance was assigned to the orientation component.

Despite not measuring the orientation directly, the EKF was able to infer it from the motion model. Over time, the estimate converged to the true pose.

7 Discussion

The most important parameters of the EKF are the process noise covariance R and the measurement noise covariance Q. It was observed that:

- Small process noise leads to smooth but less adaptive estimates.
- Larger process noise improves responsiveness but increases noise in the estimate.
- Accurate estimation of GNSS covariance significantly improves overall performance.

A major challenge was handling the nonlinear motion model and achieving stable convergence, especially when the initial pose was uncertain.

Even in the known initial pose scenario, a noticeable deviation from the ideal trajectory was observed at the beginning of the motion. This behavior is caused by the high confidence in the initial estimate. Due to the low initial covariance, the filter assumes that the predicted state is highly accurate and therefore does not sufficiently compensate for measurement noise. As a result, even a small error in orientation leads to a visible deviation in the trajectory due to the nonlinear motion model and the fact that control is based on the EKF estimate.

In contrast, the unknown initial pose scenario starts with high uncertainty, which results in smoother convergence. The filter gradually incorporates GNSS measurements and avoids aggressive corrections. Consequently, the trajectory appears more stable, even though the initial estimate is less accurate.

8 Conclusion

The implemented EKF successfully localized the robot using GNSS measurements. The filter was able to estimate both position and orientation, even though orientation was not directly measured.

The results demonstrate that the EKF provides a smooth and reliable estimate, making it suitable for real-world localization tasks.